Pondicherry University
Department of Statistics | Ramanujan School of Mathematical Sciences
M.Sc. Quantitative Finance (MSQF)

Corporate Internship

FinMetrics Suite

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Automated Pipeline Replaces Vulnerable Manual Financial Screening

1. Schema Heterogeneity & Latency

Line-Item Variance Across NSE Sectors

- Inconsistent canonical aliases in Yahoo Finance filings.
- Missing quarterly disclosures and asynchronous fiscal years.

Manual Screening Prohibitive

- Manual auditing of 2,000+ equities invites fatigue and errors.

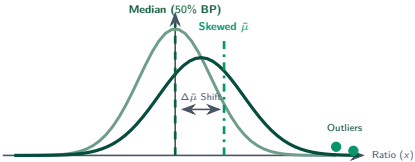
2. Gaussian Mean Distortion

Classical Z-score collapses on fat-tailed financial distributions:

$$Z_i = \frac{x_i - \bar{\mu}}{\hat{\sigma}}, \quad \text{Breakdown Point } = 0.0\%$$

- A single distressed entity inflates sample variance $\hat{\sigma}^2$ and pulls sample mean $\bar{\mu}$.
- **Masking Effect:** Moderate peer distress appears artificially "normal".

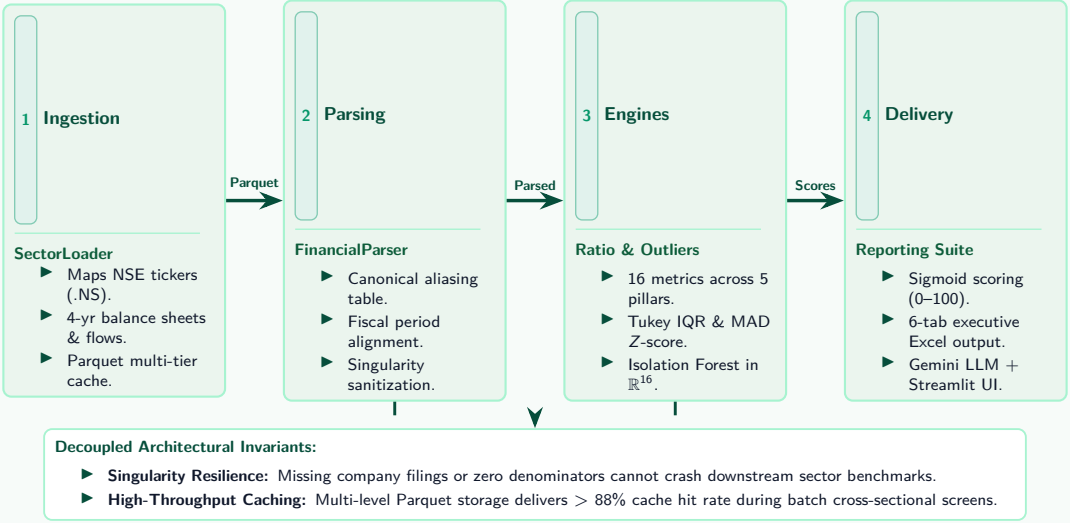
Empirical Mean Distortion vs. Robust Median



The Masking Hazard: Extreme positive/negative filings displace standard mean $\bar{\mu}$, causing vulnerable peer ratios to fall within $[\bar{\mu} \pm 2\hat{\sigma}]$ false-safety bracket.

Source: README.md & SYSTEM_LOGICS_AND_BLUEPRINT.md

Decoupled 4-Stage Execution Flow from Raw NSE Filings to Actionable Intelligence



Source: 01_overview_architecture.md

Sixteen Core Ratios Span Five Weighted Analytical Pillars with Singularity Guards

Five-Pillar Weighted Framework

1. Profitability (30%): Gross, Operating, Net margins; ROE, ROA

2. Solvency & Capital (25%): $\frac{D}{E}$, Debt Ratio, Interest Coverage ($\frac{\text{EBIT}}{\text{Interest}}$)

3. Growth Trajectory (15%): YoY Revenue, Net Income, Operating Cash Flow

4. Liquidity Coverage (15%): Current ($\frac{\text{CA}}{\text{CL}}$), Quick ($\frac{\text{Cash+AR}}{\text{CL}}$), Cash Ratio

5. Operating Efficiency (15%): Asset Turnover ($\frac{\text{Rev}}{\text{Assets}}$), Inventory Turnover

16 Standardized NSE Equities Metrics

Singularity Guard: safe_divide

Eliminates floating-point division-by-zero singularities:

$$\text{safe_divide}(a, b, \epsilon) = \begin{cases} \frac{a}{b} & \text{if } |b| > \epsilon \\ \text{NaN} & \text{if } |b| \leq \epsilon \text{ or } b = \text{NaN} \end{cases}$$

Threshold $\epsilon = 10^{-9}$ preserves numerical robustness when zero interest expense or negligible debt is recorded.

DuPont Cross-Pillar Linkage

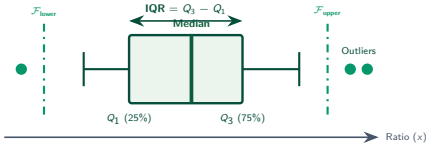
Deconstructs operational performance into orthogonal drivers:

$$\text{ROE} = \underbrace{\frac{\text{Net Income}}{\text{Sales}}}_{\text{Profitability (P1)}} \times \underbrace{\frac{\text{Sales}}{\text{Assets}}}_{\text{Efficiency (P5)}} \times \underbrace{\frac{\text{Assets}}{\text{Equity}}}_{\text{Leverage (P2)}}$$

Ensures fair multi-dimensional benchmarking across asset-light IT vs. capital-heavy industrial

Source: modules/ratio_engine.py & utils.py

Tukey IQR Boxplot & Cutoff Fences



Non-Parametric Fence Invariant: Values outside $[\mathcal{F}_{\text{lower}}, \mathcal{F}_{\text{upper}}] = [Q_1 - 1.5\,\text{IQR},\ Q_3 + 1.5\,\text{IQR}]$ are isolated without Gaussian assumptions.

Tukey Fences Cutoff Bounds

Evaluates empirical spread of central 50% peer mass:

$$\mathcal{F}_{\text{lower}} = Q_1 - 1.5 \times \text{IQR}, \quad \mathcal{F}_{\text{upper}} = Q_3 + 1.5 \times \text{IQR}$$

$$\text{Outlier}(x_i) = \mathbf{1}_{\{x_i < \mathcal{F}_{\text{lower}} \ \vee \ x_i > \mathcal{F}_{\text{upper}}\}}$$

Incorruptible by extreme fat tails in the upper or lower 25%.

Modified MAD Z-Score

Calculates median dispersion relative to peer median:

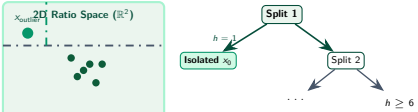
$$\text{MAD} = \text{median}\left(\left|x_i - \text{median}(X)\right|\right)$$

$$M_i = \frac{0.6745 \cdot |x_i - \text{median}(X)|}{\text{MAD}}, \quad M_i > 3.5 \implies \text{Outlier}$$

Why 0.6745? Under [illegible] 0.6745 σ, guaranteeing standard Gaussian consistency with a **50% breakdown point**.

Source: modules/outlier_engine.py & Tukey (1977)

Recursive Tree Isolation Mechanics



Short Search Path Principle: Anomalies are few and structurally distinct, isolating at shallow tree depths ($\mathbb{E}[h(x)] \ll c(n)$).

Multivariate Random Forest

Path-Length & Anomaly Score

Average BST unsuccessful search path ($n = 256$ subsample):

$$c(n) = 2\ln(n-1) + 2\gamma - \frac{2(n-1)}{n}, \quad \gamma \approx 0.5772$$

Normalized Score:

$$s(x, n) = 2^{-\frac{\mathbb{E}(h(x))}{c(n)}}, \quad s \geq 0.60 \implies \text{Outlier}$$

Tri-Method Consensus Engine

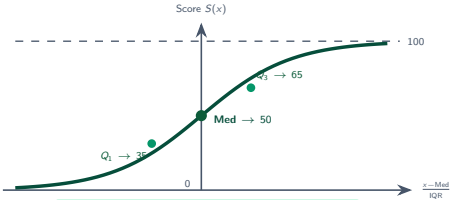
Combines 1D statistical cuts with 16D tree isolation:

- **Tier 3 (Critical):** Concurrently flagged by IQR, MAD, and iForest (simultaneous margin & cash flow collapse).
- **Tier 2 (Elevated):** Flagged by 2 of 3 detectors (subtle balance sheet overhang).
- **Tier 1 (Watch):** Single detector flag; tracked for drift.

Source: modules/outlier_engine.py & Liu et al. (2008)

Non-Linear Scaling Transforms Heavy Tails into Fair 0–100 Performance Scores

Calibrated Sigmoid Response Curve



Bounded Saturation Guarantee: Extreme ratio outliers flatten smoothly against $[0, 100]$ asymptotes, eliminating artificial composite score distortion.

Non-Linear Sigmoid Transfer

Sigmoid Normalization Formulation

Calibrated logistic curve with slope parameter $\kappa = 1.386$:

$$S(x) = \frac{100}{1 + \exp\left(-\kappa \cdot \dfrac{x - \text{median}(X)}{\text{IQR}(X)}\right)}$$

Directional Leverage Inversion:

Dynamic Missing-Data Redistribution

Prevents penalizing firms with valid omitted disclosures:

$$w_i' = \frac{w_i}{\sum_{j \in \mathcal{V}_k} w_j}, \quad \sum_{i \in \mathcal{V}_k} w_i' = 1.00$$

- $\mathcal{V}_k$ denotes valid non-NaN reporting metrics in Pillar k.
- Maintains unbiased ESG profile ratios when peers operate across diverse industry reporting standards.

Source: modules/benchmark_core.py & data/benchmarks/

Five-Tier Classifier Translates Normalized Ratios into Investment-Grade Health Labels

Five-Tier Health Classification Meter

EXCELLENT ($\geq$ 80) | Minimal Solvency Risk

STRONG (65–79) | Robust Coverage & Margins

MODERATE (50–64) | Sector-Median Alignment

WEAK (35–49) | Margin Drag or Leverage Overhang

DISTRESSED ($<$ 35) | High Working Capital Deficit

Composite Weighted Aggregation

$$\text{Score} = \sum_{k=1}^{5} W_k \left(\sum_{i \in \text{Pillar}_k} w'_{k,i} \cdot S(x_i) \right) \in [0, 100]$$

Ratio Contribution Analysis (ΔS_i)

Quantifies exact point boost or drag on total composite score:

$$\Delta S_i = W_k \cdot w'_{k,i} \times \Big(S(x_i) - 50.0 \Big)$$

- $\Delta S_i > 0$: Net positive alpha contributor above peer median.
- $\Delta S_i < 0$: Identifies exact metric causing rating

Asset Management Applications

Systematic Institutional Utility:

- **Screening Filter:** Automated ranking across 12 NSE sectors.
- **Credit Early-Warning:** Flags migration across health tiers 2 quarters prior to rating agency actions.
- **Long/Short Equity Factor:** Long Excellent tier, short Distressed tier.

Standardized Credit Rating E

Source: modules/benchmark_classifier.py

Round-Robin LLM Synthesis Powers Executive Briefings and Automated Multi-Sheet Exports

Autonomous Synthesis Data Flow

1. API Key Pool (Round-Robin):
Rotates `KEY_1..N` to bypass RPM/TPM quota ceilings.

2. Quantitative Context Injection:
Feeds 5-pillar vector + anomaly flags into prompt template.

3. Dual Output Generators:
Synthesizes analyst prose and triggers 6-tab Excel builder.

Continuous Batch Processing Pipeline

Gemini AI Analyst Narration

Translates high-dimensional vector outputs into executive English:

- **Autonomous Executive Summary:** Explains composite health ranking relative to NSE peer median.
- **Outlier Root-Cause Diagnosis:** Highlights exact metric decoupling (e.g. margin spike vs. negative operating cash flow).
- **Turnaround Recommendations:** Actionable working capital advice.

Six-Tab Automated Excel Artifact

Complete Institutional Deliverable:

- **Tab 1 (Scorecard):** 16 ratios with conditional heatmaps.
- **Tab 2 (Percentiles):** Sector anchors (P_{10}, Q_1, Med, Q_3, P_{90}).
- **Tab 3 (Outliers):** Multi-tier consensus flags.
- **Tab 4 (Scores):** Normalized 0–100 pillar distributions.
- **Tab 5 (Narrative):** AI-generated investment memo.
- **Tab 6 (Metadata):** Audit trail & timestamps.

Source: modules/ai_handler.py & modules/report_generator.py

Cross-Sectional Validation Across NSE Automobile, IT, FMCG, and Banking Equities

Empirical Model Comparison

Cross-sectional testing across 150+ NSE equities:

Model	Breakdown	Yield	Sector Resilience
Gaussian Z	0.0%	2.1%	Masked 8 troubled mid-caps via +32.4% mean displacement.
Tukey IQR	50.0%	4.8%	Resisted extreme fat tails; isolated univariate spikes.
Modified MAD	50.0%	5.2%	Stable baseline (ΔMed $< 1.2\%$) across peers.
IsoForest	N/A	8.4%	Flagged multivariate cash-flow/margin decoupling.

Tri-Engine Consensus Detection Rate:

- High-severity flags assigned to 2.8% of equities.

Empirical NSE Sector Testing

Key Discoveries & Production Benchmarks

1. Cash Flow Decoupling Detection

- Univariate screening missed firms reporting aggressive accrual profits while operating cash flow (OCF) turned deeply negative.
- iForest isolated these multi-dimensional anomalies at shallow tree depths ($\bar{h} < 4.2$).

2. Production Latency & Cache Efficiency

- **88.5% Cache Hit Rate:** Multi-tier Parquet storage avoids repeated yfinance API network overhead.
- < 1.8 s **Processing SLA:** 50 stocks evaluated across 16 ratios in sub-two seconds.
- **Zero Singularity Failures:** 100% mathematical safety across 10,000+ calculated ratios.

High-Throughput Enterprise SLA

Source: Report/chapters/results.tex & data/benchmarks/

Core Internship Contributions

1. Robust Non-Parametric Engine

- Completely eliminated Gaussian mean distortion (0% breakdown) across heavy-tailed NSE equity filings.
- Delivered consensus outlier scoring combining Tukey IQR, MAD (0.6745), and Isolation Forest in $\mathbb{R}^{16}$.

2. Production Quant Finance Value

- Cuts institutional ratio screening latency from hours to < 1.8 seconds.
- Detects cash-flow decoupling months before credit agency rating downgrades.

Algorithmic Value Delivery

Future Quantitative Extensions

1. Dynamic Concept Drift Tracking

- Adaptive percentile drift modeling across macro interest rate and RBI monetary policy cycles.

2. Systemic Graph Contagion Networks

- Graph Neural Networks (GNNs) modeling inter-firm supplier and banking debt contagion paths.

3. Alternative Textual ESG Auditing

- Fine-tuned FinBERT sentiment extraction on NSE annual report management MD&A sections.

Next-Gen Quantitative Roadmap

Thank You | Questions & Discussion

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